DT081A Signal and Image Processing

Exercise 5

Light and Spectra

5.1 Explain how light is characterized by its wavelength.
5.2 What is the wavelength range that humans can perceive?
5.3 Calculate the frequency of light with a wavelength of 500 nm. Use the speed of light in a vacuum $c = 3 \times 10^8$ m/s.
5.4 Compare and contrast a laser as a light source versus most other light sources in terms of wavelength composition.
5.5 Describe the process of image formation, from light reflection off surfaces to capture by an imaging device. Include the role of the reflectance function in this process.

Human Eye and Color Perception

5.6 Describe the difference between rods and cones in the human eye. How do their roles relate to day and night vision?
5.7 What are the three types of cones, and what part of the color spectrum does each type respond to?
5.8 The human eye has different sensitivity curves for red, green, and blue light. Given the sensitivity function for the green cone peaks at 550 nm, compute the corresponding frequency in Hz.
5.9 Define luminance and chrominance. How are these related to human vision?

Image Formation

5.10 Explain how the reflectance function of a surface influences the light captured by an imaging device.
5.11 Given the following reflectance function for a surface:

$$R(\lambda) = \frac{1}{2}\sin\left(\frac{2\pi\lambda}{700}\right) + 0.5$$

Where λ is the wavelength of light (in nm),
a) calculate the amount of light reflected at $\lambda = 550$ nm.
b) calculate the average reflectance over the visible spectrum (380-750 nm)
5.12 An image

$$I(x, y)$$

undergoes the following transformation:

$$I'(x, y) = aI(x, y) + b$$

Prove that this transformation preserves the relative ordering of pixel intensities.

Spectral Sensitivity and Image Capture

5.13 What is the luminous-efficiency sensitivity function, and how does it relate to human vision?
5.14 Compare the spectral sensitivity of the human eye's red, green, and blue cones.
5.15 Explain how a digital camera captures color images using a Bayer filter.
5.16 What is gamma correction, and why is it applied to digital images?

Color Models

5.17 Explain the RGB color model and how colors are represented in this model.
5.18 Explain the YCbCr color model. How does it separate luminance from chrominance information?
5.19 Convert the following RGB values to YCbCr:

$$R = 150, G = 75, B = 0$$

5.20 Given the following YCbCr values, convert them back to RGB:

$$Y = 100, Cb = 50, Cr = 25$$

5.21 Explain the difference between additive and subtractive color models. Provide examples of each.

Image as a Function

5.22 Define an image as a two-dimensional function.
5.23 An image is represented as a matrix of pixel values. For a 3x3 grayscale image, given by the matrix:

$$\mathbf{I} = \begin{bmatrix} 50 & 75 & 100 \\ 125 & 150 & 175 \\ 200 & 225 & 250 \end{bmatrix}$$

Apply a linear transformation that scales the pixel values by a factor of 0.8.
5.24 For a grayscale image of size 1024x768 with 8-bit depth, calculate:
a) The total number of bits required to store the image

b) The number of distinct gray levels that can be represented
5.25 Explain the concept of dithering in image processing. How does it trade spatial resolution for increased perceived color depth?

Sampling and Quantization

5.26 Explain the concepts of sampling and quantization in digital imaging.
5.27 Given a continuous image function, describe the steps required to convert it to a discrete digital image.
5.28 What are the effects of decreasing the bit depth of an image from 8 bits to 4 bits?
5.29 Compute the number of quantization levels for a 6-bit grayscale image.

Discrete Cosine Transform (DCT)

5.30 Write the 2D DCT formula for an 8x8 block of image pixels.
5.31 For the following 2x2 image block, compute the DCT:

$$\mathbf{I} = \begin{bmatrix} 100 & 120 \\ 130 & 150 \end{bmatrix}$$

5.32 Explain why the DCT is preferred over the Fourier Transform for image compression in formats like JPEG.

Fourier Transform in Image Processing

5.33 Write the formula for the 2D discrete Fourier transform (DFT).
5.34 Explain the relationship between the amplitude spectrum and the phase spectrum of an image's Fourier transform.
5.35 Consider an image with vertical stripes. Describe how the stripes would appear in the frequency domain after applying the Fourier transform.
5.36 Compute the 2D Fourier transform for the following 2x2 image block:

$$\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

5.37 Explain how to reconstruct an image from its Fourier transform.
5.38 If an image's high-frequency components are removed, what changes would you expect in the reconstructed image?

Lossy Compression

5.39 Describe the process of quantizing DCT coefficients in JPEG compression.
5.40 After applying quantization, which DCT coefficients are typically most affected, and why?
5.41 Apply a basic quantization matrix to the DCT coefficients computed in 5.28 and analyze the result.
5.42 What is the main trade-off between lossy and lossless image compression?
5.43 JPEG is a widely used lossy compression format. Describe how JPEG compresses an image using DCT and quantization.

Lossless Compression

5.44 Explain how PNG achieves lossless compression.
5.45 Compare the compression ratios of JPEG and PNG for the same image. What accounts for the difference?
5.46 Describe the principle of predictive coding in lossless JPEG compression.

Elementary Image Expansion

5.47 Explain how an image can be expressed as a linear combination of elementary images using the outer product of vectors.
5.48 Expand the following 2x2 image into elementary images:

$$\mathbf{I} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Basis Transformations

5.49 What are the advantages of using the DCT for image compression compared to using the Fourier transform?
5.50 Apply the Fourier transform to a small 4x4 image and compare the result with applying the DCT.

Spatial Filtering

5.51 Describe how spatial filtering can be used to enhance edges in an image.
5.52 Write the 3x3 convolution kernel for a simple edge-detection filter. Apply this filter to a 3x3 image block:

$$\mathbf{I} = \begin{bmatrix} 50 & 75 & 100 \\ 125 & 150 & 175 \\ 200 & 225 & 250 \end{bmatrix}$$

Smoothing and Sharpening

5.53 Explain the difference between smoothing and sharpening filters. Provide examples of each.
5.54 Given a 3x3 Gaussian smoothing filter,

$$G = \frac{1}{16}\begin{bmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 1\end{bmatrix}$$

convolve it with the following image block:

$$\mathbf{I} = \begin{bmatrix}100 & 120 & 130\\ 110 & 115 & 125\\ 140 & 150 & 160\end{bmatrix}$$

5.55 Describe how the concepts of spatial frequency and the Nyquist sampling theorem extend to 2D image sampling. What are the implications for avoiding aliasing in digital images?